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Inventor(s)	Su; Wei et al.

Bandwidth Resize Method and Apparatus

Abstract

A size of a service container is adjusted with bandwidth (or a rate) of an optical service unit (OSU) frame. When determining that bandwidth of service data needs to be resized, a source node updates a value of a first indication in a data unit frame to indicate a bandwidth resize location, and may further determine an actual start location of bandwidth changing based on the bandwidth resize location, to map a part that is in the data unit frame and that is located before the actual start location to a service container before resizing, and map a part that is in the data unit frame and that is located after the actual start location to a service container before resizing.

Inventors: Su; Wei (Dongguan, CN), Sun; Liang (Dongguan, CN), Zheng; Shuqian (Dongguan, CN)

Applicant: Huawei Technologies Co., Ltd. (Shenzhen, CN)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a continuation of International Patent Application No. PCT/CN2023/132661 filed on Nov. 20, 2023, which claims priority to Chinese Patent Application No. 202211469162.0 filed on Nov. 22, 2022. The disclosures of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] This disclosure relates to the field of optical communication technologies, and in particular, to a bandwidth resize method and apparatus.

BACKGROUND

[0003] An optical transport network (OTN) has become a mainstream technology used in a transport network because the optical transport network has features such as high bandwidth, a large capacity, high reliability, and a low latency. The OTN is widely applied to backbone, metropolitan core, aggregation, and other networks. Currently, the OTN is further expanded to an access network. Currently, the OTN may provide a high-bandwidth transmission capability such as $n \times 1.25$ gigabits per second (Gbps) or $n \times 5$ Gbps, and may provide a low-bandwidth transmission capability, for example, an optical service unit (OSU) is used. When a service signal is transmitted, the service signal is mapped to the OSU, and then the OSU is mapped to a payload area of a flexible optical data unit (ODUflex) for transmission.

[0004] In some scenarios, a rate of an OSU may increase because of a service requirement. One solution is to resize one byte every 125 microseconds (us), but fast bandwidth resizing is not supported.

SUMMARY

[0005] Embodiments of this disclosure provide a bandwidth resize method and apparatus, to improve a bandwidth resize rate.

[0006] According to a first aspect, an embodiment of this disclosure provides a bandwidth resize method, applied to a first network node. The bandwidth resize method includes the following steps: service data is obtained. The service data is mapped to a plurality of data unit frames, where a payload area of the plurality of data unit frames is used to carry the service data. When it is determined that bandwidth of the service data needs to be resized, a first indication carried in a first data unit frame of the plurality of data unit frames is set to a specified value. When the first indication is the specified value, the first indication indicates a bandwidth resize location of the service data that is in the plurality of data unit frames. A first location at which bandwidth of the plurality of data unit frames changes from first bandwidth to second bandwidth on the first network node is determined based on the bandwidth resize location. A part that is in the plurality of data unit frames and that is located before the first location is mapped to a first service container of an OTN frame. A part that is in the plurality of data unit frames and that is located after the first location is mapped to a second service container of the OTN frame. The OTN frame is sent.

[0007] In embodiments of this disclosure, a size of a service container is adjusted with bandwidth (or a rate) of an OSU. When determining that the bandwidth of the service data needs to be resized, a source node updates a value of the first indication in the data unit frames to indicate the bandwidth resize location, and may further determine, based on the bandwidth resize location, an actual start location of bandwidth changing, to map a part that is in the data unit frames and that is located before the actual start location to a service container before resizing, and map a part that is in the data unit frames and that is located after the actual start location to a service container before resizing. In comparison with an existing case in which one byte is resized every 125 μ s, in this disclosure, the first bandwidth is resized to the second bandwidth through only one operation. This increases a bandwidth resize rate.

[0008] In a possible design, an overhead area of the first data unit frame further carries a second indication. The second indication indicates a quantity of network nodes that the plurality of data unit frames pass through. A network node that the plurality of data unit frames pass through is a network node that completes mapping of the plurality of data unit frames to the OTN frame at a service layer.

[0009] In a possible design, that a first location at which bandwidth of the plurality of data unit frames changes from first bandwidth to second bandwidth on the first network node is determined based on the bandwidth resize location includes: adjusting a value of the second indication in the first data unit frame to be accumulated by 1; and determining the first location based on the bandwidth resize location and a value of the second indication after adjustment.

[0010] In the foregoing design, an actual resize location of the bandwidth is determined based on the second indication, so that a first in, first out (FIFO) waterline that is of a current node and that is used to buffer the OSU can be adjusted. This prevents an exception caused by FIFO overflow or FIFO read empty.

[0011] In a possible design, the first location satisfies:

$$[00001] W = y + HA * n$$

[0012] W represents the first location, y represents the bandwidth resize location, HA represents the second indication, and n is a preset value.

[0013] In a possible design, n is related to a bandwidth difference between the first bandwidth and the second bandwidth. n may be a quantity of slots corresponding to the bandwidth difference.

[0014] In a possible design, that a part that is in the plurality of data unit frames and that is located before the first location is mapped to a first service container of the OTN frame, and a part that is in the plurality of data unit frames and that is located after the first location is mapped to a second service container of the OTN frame includes: determining a boundary point between the first service container and the second service container in the OTN frame based on the first location, the first bandwidth, and the second bandwidth; and based on the boundary point, mapping the part that is in the plurality of data unit frames and that is located before the first location to the first service

container of the OTN frame, and mapping the part that is in the plurality of data unit frames and that is located after the first location to the second service container of the OTN frame.

[0015] In a possible design, the boundary point satisfies:

[00002]

$Y = X + \text{celling}(\frac{(W-x) \times P}{C_m(M)})$, where $X + \text{celling}(\frac{(W-x) \times P}{C_m(M)}) < P$; or $Y = X + \text{celling}(\frac{(W-x) \times P}{C_m(M)}) - P$, where $X + \text{celling}(\frac{(W-x) \times P}{C_m(M)}) \geq P$

[0016] Y represents the boundary point, y represents the bandwidth resize location, X represents a mapping location to which the first indication in the first data unit frame is mapped and that is on the OTN frame, x represents a location of the first indication in the plurality data unit frames, W represents the first location, P represents a size of the first service container or a size of the second service container, $C_{\text{sub.m}}(M)$ represents a quantity of data blocks in a service container of the OTN frame to which the plurality of data unit frames are mapped before bandwidth resizing, and $\text{celling}()$ represents rounding up.

[0017] The foregoing design provides a possible effective manner of mapping an actual resize location of the bandwidth to an actual OTN location.

[0018] In a possible design, the overhead area of the OTN frame includes an overhead region corresponding to each slot in the payload area of the OTN frame. An overhead region of a last slot included in the first service container carries a mapping overhead of mapping the plurality of data unit frames to the first service container.

[0019] In a possible design, that a part that is in the plurality of data unit frames and that is located before the first location is mapped to a first service container of the OTN frame, and a part that is in the plurality of data unit frames and that is located after the first location is mapped to a second service container of the OTN frame includes:

[0020] The part that is in the plurality of data unit frames and that is located before the first location is mapped to M slots of the OTN frame, and the part that is in the plurality of data unit frames and that is located after the first location is mapped to M+N slots or M-N slots of the OTN frame. The first service container includes the M slots, and the first bandwidth corresponds to bandwidth of the M slots. The second service container includes the M+N slots or includes the M-N slots, and the second bandwidth corresponds to bandwidth of the M+N slots or bandwidth of the M-N slots.

[0021] In a possible design, the data unit frame is an OSU, and/or the OTN frame is an optical data unit (ODU) frame.

[0022] In a possible design, the first service container includes the M slots, and the first bandwidth corresponds to the bandwidth of the M slots. The second service container includes the M+N slots or includes the M-N slots, and the second bandwidth corresponds to the bandwidth of the M+N slots or the bandwidth of the M-N slots. The overhead area of the OTN frame includes the overhead region corresponding to each slot in the payload area of the OTN frame, and overhead regions corresponding to the N slots are used to carry interaction information needed for the bandwidth resizing.

[0023] In a possible design, the interaction information includes one or more of a resize protocol (RP), tributary slot connectivity check (TSCC), a tributary port identifier (TPID), control information (CTRL), and a tributary slot group status (TSGS).

[0024] In a possible design, the first network node is a source node on a transmission path of the service data.

[0025] According to a second aspect, an embodiment of this disclosure provides another bandwidth resize method, applied to a second network node. The method includes: an OTN frame is received. The OTN frame is used to carry a plurality of data unit frames, and payload areas of the plurality of data unit frames carry service data. A first data unit frame is demapped from a first service container of the OTN frame. A first indication carried in the first data unit frame is a specified value, and indicates a bandwidth resize location of the service data in the plurality of data unit frames. A first location at which bandwidth of the plurality of data unit frames changes from first bandwidth to second bandwidth is determined based on the bandwidth resize location. A part that is in the plurality of data unit frames and that is located before the first location continues to be demapped from the first service container of the OTN frame, and a part that is in the plurality of data unit frames and that is located after the first location is demapped from a second service container of the OTN frame.

[0026] In a possible design, an overhead area of the first data unit frame further carries a second indication. The second indication indicates a quantity of network nodes that the plurality of data unit frames pass through. The network node that the plurality of data unit frames pass through is a network node that completes mapping of the plurality of data unit frames to the OTN frame at a service layer.

[0027] In a possible design, that a first location at which bandwidth of the plurality of data unit frames changes from first bandwidth to second bandwidth is determined based on the bandwidth resize location includes: determining the first location based on the bandwidth resize location and a value of the second indication.

[0028] In a possible design, the first location satisfies:

[00003] $W = y + HA * n$

[0029] W represents the first location, y represents the bandwidth resize location, HA represents the second indication, and n is a preset value.

[0030] In a possible design, that a part that is in the plurality of data unit frames and that is located before the first location continues to be demapped from the first service container of the OTN frame, and a part that is in the plurality of data unit frames and that is located after the first location is demapped from a second service container of the OTN

frame includes: determining a boundary point between the first service container and the second service container in the OTN frame based on the first location, the first bandwidth, and the second bandwidth; and based on the boundary point, continuing to demap the part that is in the plurality of data unit frames and that is located before the first location from the first service container of the OTN frame, and demapping the part that is in the plurality of data unit frames and that is located after the first location from the second service container of the OTN frame.

[0031] In a possible design, the boundary point satisfies:

[00004]

$Y = X + \text{celling}(\frac{(W-x) \times P}{C_m(M)})$, where $X + \text{celling}(\frac{(W-x) \times P}{C_m(M)}) < P$; or $Y = X + \text{celling}(\frac{(W-x) \times P}{C_m(M)}) - P$, where $X + \text{celling}(\frac{(W-x) \times P}{C_m(M)}) \geq P$.

[0032] Y represents the boundary point, y represents the bandwidth resize location, X represents a mapping location to which the first indication in the first data unit frame is mapped and that is on the OTN frame, x represents a location of the first indication in the plurality data unit frames, W represents the first location, P represents a size of the first service container or a size of the second service container, C.sub.m(M) represents a quantity of data blocks in a service container of the OTN frame to which the data unit frame is mapped before bandwidth resizing, and celling() represents rounding up.

[0033] In a possible design, the overhead area of the OTN frame includes an overhead region corresponding to each slot in the payload area of the OTN frame. An overhead region of a last slot included in the first service container carries a mapping overhead of mapping the plurality of data unit frames to the first service container.

[0034] In a possible design, the data unit frame is an OSU frame, and/or the OTN frame is an ODU.

[0035] In a possible design, the first service container includes M slots, and the first bandwidth corresponds to bandwidth of the M slots. The second service container includes M+N slots or includes M-N slots, and the second bandwidth corresponds to bandwidth of the M+N slots or bandwidth of the M-N slots.

[0036] In a possible design, the overhead area of the OTN frame includes the overhead region corresponding to each slot in the payload area of the OTN frame, and overhead regions corresponding to N slots are used to carry interaction information needed for the bandwidth resizing.

[0037] In a possible design, the interaction information includes one or more of an RP, TSCC, a TPID, CTRL, and a TSGS.

[0038] In a possible design, the first network node is a sink node or an intermediate node on a transmission path of the service data.

[0039] According to a third aspect, an embodiment of this disclosure provides a bandwidth resize apparatus, used in a network node. The apparatus includes a processor and an optical transceiver. The processor is configured to: perform the method according to the first aspect or any design of the first aspect, or perform the method according to the second aspect or any design of the second aspect, and receive an OTN frame by using the optical transceiver.

[0040] According to a fourth aspect, an embodiment of this disclosure provides a bandwidth resize apparatus, used in a network node. The apparatus includes a processor and a memory. The memory is configured to store program code. The processor is configured to: read and execute the program code stored in the memory, to implement the method according to the first aspect or any design of the first aspect, or implement the method according to the second aspect or any design of the second aspect.

[0041] According to a fifth aspect, an embodiment of this disclosure further provides a computer storage medium. The storage medium stores a software program. When the software program is read and executed by one or more processors, the method provided in the first aspect or any design of the first aspect may be implemented, or the method provided in the second aspect or any design of the second aspect may be implemented.

[0042] According to a sixth aspect, an embodiment of this disclosure provides a computer program product including instructions. When the computer program product runs on a computer, the computer is enabled to perform the method provided in any design of the first aspect, or the computer is enabled to perform the method provided in any design of the second aspect.

[0043] According to a seventh aspect, an embodiment of this disclosure provides a chip, and the chip includes a processor. The processor is configured to: perform the method provided in any design of the first aspect, or perform the method provided in any design of the second aspect.

[0044] In a possible design, the chip further includes a communication interface, and the communication interface is coupled to the processor.

[0045] In a possible design, the chip is connected to the memory, and is configured to read and execute a software program stored in the memory, to implement the method provided in any design of the first aspect or implement the method provided in any design of the second aspect.

[0046] On a basis of the implementations provided in the foregoing aspects, this disclosure may further provide more implementations by combining the implementations.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0047] FIG. 1 is a schematic diagram of a network architecture.

[0048] FIG. 2 is a schematic diagram of a hardware structure of a network device.

[0049] FIG. 3 is a schematic diagram of tributary slots or time slots (TS) distribution using generic mapping procedure (GMP) mapping.

[0050] FIG. 4 is a schematic flowchart of a bandwidth resize method according to an embodiment of this disclosure.

[0051] FIG. 5 is a schematic diagram of a structure of a possible OSU according to an embodiment of this disclosure.

[0052] FIG. 6 is a schematic diagram of a service container change according to an embodiment of this disclosure.

[0053] FIG. 7 is a schematic diagram of a structure of a resize control overhead (RCOH) according to an embodiment of this disclosure.

[0054] FIG. 8 is a schematic diagram of ODU0 overhead area distribution according to an embodiment of this disclosure.

[0055] FIG. 9 is a schematic diagram of an overhead region of a last slot of a first service container according to an embodiment of this disclosure.

[0056] FIG. 10 is a schematic diagram of slot overhead regions respectively corresponding to an even multiframe and an odd multiframe according to an embodiment of this disclosure.

[0057] FIG. 11 is a schematic flowchart of a bandwidth resize method according to an embodiment of this disclosure.

[0058] FIG. 12 is a schematic diagram of a transmission path of service data according to an embodiment of this disclosure.

[0059] FIG. 13A and FIG. 13B are schematic flowcharts of a bandwidth resize method in a scenario of increasing bandwidth according to an embodiment of this disclosure.

[0060] FIG. 14 is a schematic diagram of an end-to-end bandwidth resize procedure according to an embodiment of this disclosure.

[0061] FIG. 15 is a schematic diagram of a structure of a possible bandwidth resize apparatus according to an embodiment of this disclosure.

[0062] FIG. 16 is a schematic diagram of a structure of another possible bandwidth resize apparatus according to an embodiment of this disclosure.

DESCRIPTION OF EMBODIMENTS

[0063] The following describes the technical solutions of embodiments of this disclosure with reference to the accompanying drawings in embodiments of this disclosure.

[0064] In the descriptions of this disclosure, unless otherwise specified, “a plurality of” means two or more. In addition, “/” indicates that the objects associated with each other are in an “or” relationship. For example, A/B may represent A or B. In this disclosure, the term “and/or” describes an association relationship between associated objects and indicates that three relationships may exist. For example, A and/or B may indicate the following three cases: only A exists, both A and B exist, and only B exists. A and B each may be singular or plural. In addition, to clearly describe the technical solutions in embodiments of this disclosure, in embodiments of this disclosure, words such as “first” and “second” are used to distinguish same items or similar items that have basically same functions and functions. Persons skilled in the art may understand that the terms such as “first” and “second” do not limit a quantity or an execution sequence, and the terms such as “first” and “second” do not indicate a definite difference. It should be further noted that, unless otherwise specified, specific descriptions for some technical features in one embodiment may further be used to explain corresponding technical features mentioned in other embodiments.

[0065] A network architecture and a service scenario described in embodiments of this disclosure are intended to describe technical solutions in embodiments of this disclosure more clearly, and do not constitute any limitation on the technical solutions provided in embodiments of this disclosure. Persons of ordinary skill in the art may learn that the technical solutions provided in embodiments of this disclosure are also applicable to a similar technical problem as a network architecture evolves and a new service scenario emerges.

[0066] The technical solutions provided in embodiments of this disclosure are applicable to an optical network, for example, an OTN. One OTN is usually formed by connecting a plurality of OTN devices through an optical fiber, and may have different topological types such as a line type, a ring type, and a mesh type based on a specific requirement.

[0067] FIG. 1 is a schematic diagram of a network architecture. An OTN shown in FIG. 1 includes two OTN networks (namely, an OTN network 1 and an OTN network 2). Each OTN network includes a specific quantity of network devices (which may also be referred to as OTN devices). A link between network devices in the OTN network is an intra-domain link, and a link between network devices between OTN networks is an inter-domain link. Based on an actual requirement, one network device may have one or more functions. Generally, network devices are classified into an optical layer device, an electrical layer device, and an optical-electrical hybrid device. The optical layer device is a device that can process an optical layer signal, for example, an optical amplifier (OA). The electrical layer device is a device that can process an electrical layer signal, for example, a device that can process an ODU signal. The optical-electronic hybrid device is a device capable of processing the optical layer signal and the electrical layer signal. It should be noted that, based on a specific integration requirement, one network device may integrate a

plurality of different functions. The technical solutions provided in this disclosure are applicable to network devices in different forms and integration degrees. As shown in FIG. 1, the OTN network may further be connected to a client device, for example, a first client device connected to a network device 1 or a second client device connected to a network device 3 in FIG. 1. A quantity of client devices connected to the OTN network is not limited in embodiments of this disclosure. The client device may be user equipment, or may be a server. For example, the second client device is a cloud server. An example in which the OTN network is independently connected to one server to provide a service for a user is used in FIG. 1, and an example of the server is a server connected to a network device 7 in FIG. 1. The network device in the OTN network is configured to transmit service data of the client device. For example, the network device may receive the service data of the client device, then map the service data to an OSU frame, further map the OSU frame to an OTN frame, and send the OTN frame to another network device. For example, the service data may be data of a constant bit rate (CBR) service, or data of a packet (PKT) type service, or data of a synchronous transport module (synchronous transport module-N (STM-N)) service. It should be noted that the network device in embodiments of this disclosure may also be referred to as a network node.

[0068] In some embodiments, different OTN networks may be managed by different network management devices. In FIG. 1, a network management device 1 manages the OTN network 1, and a network management device 2 manages the OTN network 2.

[0069] FIG. 2 is a schematic diagram of a hardware structure of a network device. The network device includes a power supply, a fan, and an auxiliary board, and may further include a tributary board, a line board, a cross-connect board, and a system control and communication board. The line board may also include an optical layer processing board. It should be noted that each device may include different board types and a different quantity of boards based on a specific requirement. For example, a network device serving as a core node may have no tributary board. A network device serving as an edge node may have a plurality of tributary boards. The power supply is used to supply power to the network device, and may include an active power supply and a standby power supply. The fan is used to dissipate heat of the device. The auxiliary board is used to provide an auxiliary function, for example, providing an external alarm or accessing an external clock. The tributary board, the cross-connect board, and the line board are mainly configured to process an OTN electrical layer signal (referred to as an OTN frame below). The tributary board is configured to receive and send various client services, for example, a synchronous digital hierarchy (SDH) service, a packet service, an Ethernet service, and a fronthaul service. Further, the tributary board may be divided into an optical module on a client side and a signal processor. The optical module on the client side may be an optical transceiver, and is configured to receive and/or send a client signal. The signal processor is configured to implement mapping and demapping between the client signal and the OTN frame. The cross-connect board is configured to implement exchange of the OTN frame, and complete exchange of one or more types of OTN frames. The line board mainly implements processing of an OTN frame on a line side. The line board may be divided into an optical module on the line side and a signal processor. The optical module on the line side may be an optical transceiver on the line side, and is configured to receive and/or send the OTN frame. The signal processor is configured to implement multiplexing and demultiplexing, or mapping and demapping of the OTN frame on the line side. The system control and communication board are used to implement system control and communication. Information may be collected from different boards via a backplane, or a control instruction is sent to a corresponding board via a backplane. Unless otherwise specified, there may be one or more specific components (for example, tributary boards). This is not limited in this disclosure. It should be noted that types of boards included in the device and specific function designs and quantities of the boards are not limited in embodiments of this disclosure.

[0070] The following first describes technical concepts in embodiments of this disclosure. [0071] (1) A network node may be referred to as a node for short, or may be referred to as a network device, and is, for example, an OTN device. [0072] (2) A data unit frame includes an integer number of bytes or bits. The data unit frame may also be referred to as an OSU frame, a flexible optical service unit (OSUflex), an optical service data unit frame, an optical service unit, a payload code block, a payload block, a switching unit or a switching code block, or a first OTN frame. This is not limited in this disclosure. The data unit frame may alternatively be of another frame type, and a rate of the data unit frame is lower than a rate of a mapped OTN frame. FIG. 3 is a possible schematic diagram of a structure of a data unit frame. As shown in FIG. 3, the data unit frame includes an overhead area and a payload area. Overheads may include but are not limited to at least one of the following: a service frame header indication, a trail trace identifier (TTI), an X bit-interleaved parity BIP-X (BIP-X), a backward error indication (BEI), a backward defect indication (BDI), a status indication (STAT), a timestamp, a sequence identifier, a mapping overhead, and the like. The payload area is used to carry service data. For example, the data unit frame may be 8 bytes (B) or an integer multiple of 16B, for example, 16B, 32B, 64B, 128B, 196B, 256B, or 512B. This is not limited in embodiments of this disclosure. [0073] (3) An OTN frame is used to carry various service data, and can manage and monitor the service data. The OTN frame may be an ODU k, an ODUCn, an ODUflex, an OTU k, an OTUCn, an optical payload unit (OPU), a flexible OTN (FlexO) frame, or the like. The OTN frame may alternatively be an ODU multiframe, an OTU multiframe, an OPU multiframe, or the like. The OTN frame may alternatively be another frame structure applicable to an optical network. [0074] (4) A GMP is a general mapping procedure of an OTN. A Cm value of a quantity of client entities and clock information in

each service frame periodicity is generated. Then, a “sigma-delta” algorithm is used to calculate a distribution pattern of client data in a payload area, and an amount of Cm client data is mapped to a corresponding location of the distribution pattern.

[0075] A data unit frame may be mapped to an OTN frame in a GMP mapping manner. The data unit frame occupies several slots (TSs) that are in the OTN frame.

[0076] For example, FIG. 3 is a schematic diagram of TS distribution using GMP mapping. In FIG. 3, a GMP OH represents an overhead area. A service container formed by one or more TSs in FIG. 3 is used to transmit an OSU. In FIG. 3, that the OTN frame is an ODU0 is used as an example. The ODU0 includes 119 10M slots. A payload area of the ODU0 is divided into 119 10M slots. A slot interleaved granularity is 16 bytes. In FIG. 3, each block is 16 bytes, and each slot in each ODU frame includes eight 16 bytes. 32 ODU0s form one multiframe. A size of the OSU is an integer multiple of 16 bytes. When the OSU is mapped to the service container, a bit stream of the OSU may be placed from top to bottom of rows and from left to right in each row. After corresponding slots in a row are full, a similar operation is performed in corresponding slots in a next row. As shown in FIG. 3, in an example, the service container occupies three slots, for example, a TS #1, a TS #2, and a TS #k, and a mapping granularity is 16 bytes. The mapping granularity may be referred to as one data block, in other words, a size of one data block is 16 bytes. In this case, after the first row is full in an order of the TS #1, the TS #2, the TS #k, the TS #1, the TS #2, and the TS #k, the bit stream of the OSU is continued to be placed in the second row in the foregoing order. [0077] (5) An asynchronous mapping procedure (AMP) is a mapping procedure of an OTN. In the AMP, a value of a positive justification byte (positive justification opportunity (PJO)) or a value of a negative justification byte (negative justification opportunity (NJO)) is determined based on a rate of a client side signal, to determine that content in a justification byte is a padding byte or a client signal, and enable a rate of a server layer signal to match a rate of a client layer signal.

[0078] Packet service traffic changes irregularly. Therefore, different traffic requirements are met through bandwidth changing of an OSU. In a case of resizing bandwidth of the OSU, resizing of bandwidth, namely, resizing of slots that are in an OTN frame, occupied when the OTN frame carries the OSU is involved. The service container is a space that carries the OSU, and the service container includes a plurality of slots that are in the OTN frame.

[0079] In a current lossless bandwidth resize solution (based on a G.HAO protocol), a periodic latency indication is used. As a result, fast bandwidth resizing cannot be implemented. A manner used in a GMP is to adjust one byte every 125 μ s, but an adjustment rate is slow in this manner.

[0080] Embodiments of this disclosure provide a bandwidth resize method and apparatus. In a bandwidth resize process, a size of a service container is synchronously adjusted as a rate (or bandwidth) of a data unit frame changes. In embodiments of this disclosure, an example in which the data unit frame is an OSU frame is used for description. For ease of description, the OSU frame is referred to as an OSU for short below. The following uses an example in which bandwidth of the OSU changes from first bandwidth to second bandwidth. For example, in a scenario of increasing the bandwidth, the first bandwidth is less than the second bandwidth. In a scenario of decreasing the bandwidth, the first bandwidth is greater than the second bandwidth.

[0081] For example, in the scenario of increasing the bandwidth, the first bandwidth corresponds to bandwidth of M slots. The second bandwidth corresponds to bandwidth of M+N slots. In the scenario of decreasing the bandwidth, the first bandwidth corresponds to bandwidth of M slots, and the second bandwidth corresponds to bandwidth of M-N slots. Both M and N are positive integers.

[0082] For ease of description, in embodiments of this disclosure, a service container corresponding to the OSU that carries the first bandwidth is referred to as a first service container, and a service container corresponding to the OSU that carries the second bandwidth is referred to as a second service container. In some scenarios, the size of the service container is synchronously adjusted when the bandwidth is resized. For example, the first service container includes the M slots, in the scenario of increasing the bandwidth, the second service container includes the M+N slots, and in the scenario of decreasing the bandwidth, the second service container includes the M-N slots. The first service container may also be understood as an optical service tributary unit.M (OSTU.M), where M represents a quantity of slots. The second service container may be understood as an OSTU.M+N or an OSTU.M-N.

[0083] In other scenarios, the size of the service container may be first increased before the bandwidth is increased, or the size of the service container is decreased after the bandwidth is decreased. In other words, the first service container and the second service container may be understood as an OSTU.M+N or an OSTU.M-N.

[0084] In embodiments of this disclosure, in a bandwidth resize process, to enable each network node to understand a bandwidth resize location that is in the OSU, it is proposed that a first indication is added to the OSU. The first indication indicates the bandwidth resize location that is in the OSU. The first indication may also be referred to as a rate change indication, a bandwidth change indication, or a bandwidth resize indicate (BWR_IND), or may be named by another name. This is not limited in embodiments of this disclosure.

[0085] The following first describes a mapping process based on the first indication of each network node on a transmission path.

[0086] FIG. 4 is a schematic flowchart of a bandwidth resize method according to an embodiment of this disclosure. The bandwidth resize method is applicable to a source node. The source node may also be referred to as the 1.sup.st

node. [0087] **401:** Obtain service data. [0088] **402:** Map the service data to a plurality of OSUs, where payload areas of the plurality of OSUs are used to carry the service data.

[0089] Frame overheads of the plurality of OSUs include a first indication. When bandwidth of the service data needs to be resized, bandwidth of the OSU needs to be resized, and this may also be understood as that a rate of the OSU needs to be adjusted. When the bandwidth of the OSU needs to be resized, or before the rate of the OSU changes, the first indication carried in a first OSU of the plurality of OSUs is set to a specified value. When the first indication is the specified value, the first indication indicates a bandwidth resize location that is of the service data and that is in the plurality of OSUs, in other words, indicates a rate adjustment location that is in the plurality of OSUs.

[0090] In an example, when a set bit of the first indication changes from 0 to 1, it indicates that the bandwidth needs to be resized. Another bit of the first indication indicates the bandwidth resize location.

[0091] In another example, when a value of the first indication changes from 0 to 1, it indicates that the bandwidth needs to be resized. A distance between a bit on which the first indication is located and the bandwidth resize location is a specific distance. Determining of the specific distance is related to a structure of the OSU. For example, different specific distances are configured for different structures of OSUs. Therefore, the bandwidth resize location may be determined based on a location that is of the first indication and that is in the plurality of OSUs and the structure of the OSU.

[0092] On this basis, **403** may be performed after **402** is performed. [0093] **403:** When it is determined that the bandwidth of the service data needs to be resized, set the first indication carried in the first OSU of the plurality of OSUs to the specified value, and when the first indication is the specified value, indicate the bandwidth resize location that is of the service data and that is in the plurality of OSUs.

[0094] It should be noted that the location that is of the first indication and that is in the plurality of OSUs is located before the bandwidth resize location.

[0095] A size of the OSU may be a multiple of 16 bytes. For example, FIG. 5 is a schematic diagram of a possible structure of an OSU according to an embodiment of this disclosure. In FIG. 5, an OSU with four rows and 960-byte columns is used as an example. The OSU includes an overhead area and a payload area. The overhead area of the OSU includes a first indication. For example, the first indication may be classified as an RCOH. The RCOH is located in an overhead region of the OSU, for example, four high bits of the eighth byte column in the first to third rows. A distance between a location of the RCOH in an OSU and a start location of a next OSU is 120×16 bytes. For example, the first indication in the overhead area of the OSU may indicate that bandwidth resizing starts to be performed on the next OSU. A specific size of the OSU and a location of the resize control overheads are not restricted. Locations of resize control overheads in OSUs of different sizes may be different. For the different sizes of the OSUs, a distance between a location x of the first indication and a bandwidth resize location y indicated by the first indication may be different.

[0096] **404:** Determine, based on the bandwidth resize location, a first location at which bandwidth of the plurality of OSUs changes from first bandwidth to second bandwidth on a first network node. [0097] **405:** Map a part that is in the plurality of OSUs and that is located before the first location to a first service container of an OTN frame, and map a part that is in the plurality of OSUs and that is located after the first location to a second service container of the OTN frame. [0098] **406:** Send the OTN frame.

[0099] For example, the “map a part that is in the plurality of OSUs and that is located before the first location to a first service container of an OTN frame, and map a part that is in the plurality of OSUs and that is located after the first location to a second service container of the OTN frame” may be implemented in the following manners.

[0100] The part that is in the plurality of OSUs and that is located before the first location is mapped to M slots of the OTN frame, and the part that is in the plurality of OSUs and that is located after the first location is mapped to $M+N$ slots or $M-N$ slots of the OTN frame.

[0101] The first service container includes at least the M slots, the first bandwidth corresponds to bandwidth of the M slots. The second service container includes the $M+N$ slots or includes the $M-N$ slots, and the second bandwidth corresponds to bandwidth of the $M+N$ slots or bandwidth of the $M-N$ slots.

[0102] In a possible implementation, in a scenario of increasing the bandwidth, in other words, when the first bandwidth is less than the second bandwidth, before the bandwidth (or the rate) of the OSU is resized (adjusted), each network node may be first indicated to increase a size of a service container. To be specific, the size of the service container increases from the M slots to the $M+N$ slots. Then, network nodes starting from the source node gradually resize the bandwidth of the OSU in a path sequence. In this case, when the first indication is received, bandwidth of the service container is bandwidth obtained through the bandwidth increasing. Further, after the first location is determined, the part that is in the plurality of OSUs and that is located before the first location is mapped to a service container (occupying the $M+N$ slots) obtained through size adjustment of the OTN frame, and the part that is in the plurality of OSUs and that is located after the first location is mapped to a service container (occupying the $M+N$ slots) obtained through size adjustment of the OTN frame.

[0103] In an example, in a scenario of decreasing the bandwidth, when the first bandwidth is greater than the second bandwidth, network nodes starting from the source node may start to gradually decrease the bandwidth of the OSU in a path sequence, and then each network node is indicated to decrease the size of the service container. In this case,

when the first indication is received, the bandwidth of the service container is still the first bandwidth. Subsequently, the source node may indicate each network node to decrease the size of the service container. Therefore, after the first location is determined, the part that is in the plurality of OSUs and that is located before the first location is mapped to the M slots of the OTN frame, and the part that is in the plurality of OSUs and that is located after the first location is mapped to the M slots of the OTN frame. Then, a specific indication is performed, and network nodes starting from the source node map an OSU received after the specific indication to the M-N slots of the OTN frame.

[0104] In another possible implementation, bandwidth of the first service container is the first bandwidth, and bandwidth of the second service container is the second bandwidth. In this implementation, the size of the service container is synchronously adjusted when the bandwidth (a transmission rate) of the OSU is resized (adjusted).

[0105] In an example, the OTN frame is an ODU frame. FIG. 6 is a schematic diagram of a service container change according to an embodiment of this disclosure. For example, an ODU0 includes 119 10M slots. 32 ODU0 frames form one multiframe. A payload area of the multiframe is divided into the 119 10M slots. A slot interleaved granularity is 16 bytes. In FIG. 6, each block is 16 bytes, and each slot includes 256×16 bytes.

[0106] A first service container OSTU.M includes M slots, a size of the first service container OSTU.M is represented by PM, and the first service container OSTU.M includes M*256×16 bytes. A second service container OSTU.M+N includes M+N slots, and a size of the second service container OSTU.M+N is represented by PM+N. The second service container OSTU.M+N includes (M+N)*256×16 bytes. The first service container occupies a TS #1, a TS #2, and a TS #k. The second service container occupies the TS #1, the TS #2, the TS #k, a TS #i, a TS #j, and a TS #119. A filling pattern part of dots in FIG. 6 corresponds to the first service container. When the OSU is mapped to the first service container, a rate of the OSU is R.sub.M, which represents M 10M bit rates. A filling pattern part of slashes in FIG. 6 corresponds to the second service container. When the OSU is mapped to the second service container, a rate of the OSU is R.sub.M+N, which represents (M+N) 10M bit rates.

[0107] A GMP mapping manner is used as an example. Before bandwidth changes, the OSU is evenly mapped to the first service container based on C.sub.m(M) information and a sigma-delta algorithm by using a GMP. The size of the first service container is the first (Y-1) 16 bytes of the OSTU.M. C.sub.m(M) represents a quantity of 16 bytes to which the OSU needs to be mapped and that is on the OSTU.M. Y represents a mapping location that corresponds to the first location and that is in a service container, namely, a boundary point between the first service container and the second service container in the OTN frame.

[0108] After the bandwidth changes, the OSU is evenly mapped to the second service container based on C.sub.m(M+N) information and the sigma-delta algorithm by using the GMP. When the bandwidth changes, the size of the second service container is the last Z 16 bytes of the OSTU.M+N. C.sub.m(M+N) represents a quantity of 16 bytes to which the OSU needs to be mapped and that is on the OSTU.M+N.

[0109] In a possible implementation, when the bandwidth is increased, each network node on a transmission path needs to complete the bandwidth increase in sequence from a source node to a sink node. As a result, a larger quantity of upstream network nodes of a node leads to a longer delay of completing the bandwidth increase on the node. In embodiments of this disclosure, a quantity of upstream network nodes of each network node is indicated based on a second indication before bandwidth resizing. The second indication may also be referred to as hop accumulation (HOPACC), or may be named by another name. This is not limited in embodiments of this disclosure. The second indication indicates a quantity of network nodes that the OSU passes through. The network node that the OSU passes through is a network node that completes mapping of a plurality of data unit frames to the OTN frame at a service layer. For example, the second indication is carried in an overhead area of a first OSU. For example, the second indication is carried in an RCOH of the first OSU. FIG. 7 is a schematic diagram of a structure of an RCOH in an OSU frame according to an embodiment of this disclosure. In some scenarios, the HOPACC may alternatively not be transmitted.

[0110] Before mapping at the service layer, each network node first increases a value of the second indication HOPACC by 1. A purpose of increasing the HOPACC by 1 is to adjust a FIFO apparatus waterline for buffering OSU data on a current network node. Generally, more occupied slots indicate a higher specified waterline. To ensure uninterrupted transmission of a bit stream, a FIFO apparatus needs to be deployed on the network node to buffer the OSU. For example, the FIFO apparatus may use a buffer manner. When each network node determines, before performing mapping, the first location at which the rate of the OSU actually changes, the network node delays n data block sizes relative to the location at which the input bit stream changes. FIG. 6 is used as an example, and a data block size is 16 bytes. Therefore, delayed n 16-byte OSU data is buffered in the FIFO apparatus, to adjust the FIFO waterline of the current network node.

[0111] In this implementation, when the first location is determined based on the first indication, the value of the second indication may be first accumulated by 1, and then the first location is determined based on the bandwidth resize location indicated by the first indication and a value of the adjusted second indication.

[0112] For example, the first location meets a condition shown in the following formula (1).

[00005]
$$W = y + HA * n \quad \text{Formula(1)}$$

[0113] W represents the first location, y represents the bandwidth resize location indicated by the first indication, HA

represents the value of the second indication, and n is related to a preset value. For example, n is related to a bandwidth difference between the first bandwidth and the second bandwidth. For example, n may be equal to a difference between a quantity of slots corresponding to the second bandwidth and a quantity of slots corresponding to the first bandwidth. In some scenarios, n may also be equal to 0.

[0114] In some scenarios, when the HOPACC does not need to be transmitted, it may be understood that the bandwidth resize location indicated by the first indication is used as the first location, that is, $W=y$.

[0115] The following describes a manner of determining the boundary point between the first service container and the second service container. The boundary point may also be understood as a mapping location to which a rate change location (namely, the first location) of the OSU is mapped and that is on the OTN frame. For example, the boundary point between the first service container and the second service container in the OTN frame may be determined based on the first location, the first bandwidth, and the second bandwidth. Therefore, the part that is in the plurality of OSUs and that is located before the first location is mapped to the first service container of the OTN frame based on the boundary point, and the part that is in the plurality of OSUs and that is located after the first location is mapped to the second service container of the OTN frame based on the boundary point.

[0116] For example, the boundary point meets a condition shown in the following formula (2) or formula (3).

$$[00006] Y = X + \text{ceiling}\left(\frac{(W-x) \times P}{C_m(M)}\right), \text{ where } X + \text{ceiling}\left(\frac{(W-x) \times P}{C_m(M)}\right) < P \quad \text{Formula(2)}$$

$$Y = X + \text{ceiling}\left(\frac{(W-x) \times P}{C_m(M)}\right) - P, \text{ where } X + \text{ceiling}\left(\frac{(W-x) \times P}{C_m(M)}\right) \geq P \quad \text{Formula(3)}$$

[0117] Y represents the boundary point, X represents a mapping location to which the first indication in the first OSU is mapped and that is on the OTN frame, x represents a location of the first indication in the first OSU, W represents the first location, P represents a size of the first service container or a size of the second service container, $C_{\text{sub.m}}(M)$ represents a quantity of data blocks of the OSU before the bandwidth resizing, where the OSU is mapped to a service container of the OTN frame, and ceiling () represents rounding up.

[0118] The formula (1), the formula (2), and the formula (3) may be combined and transformed into a formula (4) and a formula (5).

$$[00007] Y = X + \text{ceiling}\left(\frac{(y + HA * n - x) \times P}{C_m(M)}\right), \text{ where } X + \text{ceiling}\left(\frac{(y + HA * n - x) \times P}{C_m(M)}\right) < P \quad \text{Formula(4)}$$

$$Y = X + \text{ceiling}\left(\frac{(y + HA * n - x) \times P}{C_m(M)}\right) - P, \text{ where } X + \text{ceiling}\left(\frac{(y + HA * n - x) \times P}{C_m(M)}\right) \geq P \quad \text{Formula(5)}$$

[0119] For example, the example in FIG. 6 is still used, and P may be equal to $P_{\text{sub.M}}$ or $P_{\text{sub.M+N}}$. $P_{\text{sub.M}} = M * 256$, and $P_{\text{sub.M+N}} = (M+N) * 256$.

[0120] After a location of the boundary point is determined, if the size of the second service container $OSTU.M+N$ in the OTN frame is represented as Z when the bandwidth changes, Z meets a condition shown in the following formula (6):

$$[00008] Z = P_{M+N} + \text{floor}\left(\frac{Y-1}{M}\right) * (M+N) - Y \% M \quad \text{Formula(6)}$$

[0121] floor () represents rounding down, and M+N represents a quantity of slots occupied by the second service container in the scenario of increasing the bandwidth.

[0122] With reference to the example shown in FIG. 6, when the bandwidth changes, the second service container $OSTU.M+N$ in the OTN frame is of a range of a bold black-line box. To be specific, the size of the second service container $OSTU.M+N$ in the OTN frame is Z data blocks.

[0123] Embodiments of this disclosure are applicable to both GMP mapping and asynchronous AMP mapping. For example, the foregoing formula (2) to formula (5) are applicable to the GMP mapping. A principle of determining the boundary point when the AMP mapping is used is similar to a principle of determining the boundary point when the GMP mapping is used. When the AMP is used, when the boundary point may be determined, a quantity of fixed padding data blocks that needs to be inserted when the OSU is mapped to the service container may be counted based on $y-x$ or $y+HOPACC*n-x$. Therefore, proper deformation is performed based on the foregoing formula (2) to formula (5).

[0124] In some possible scenarios, some overhead information is transmitted via the OTN frame. An overhead area of the OTN frame includes an overhead region corresponding to each slot in a payload area of the OTN frame.

[0125] For example, the OTN frame is an ODU0. 32 ODU0s form one multiframe. A payload area of the ODU0 is divided into 119 10M slots, and the slot interleaved granularity is 16 bytes. FIG. 8 is a schematic diagram of ODU0 overhead area distribution according to an embodiment of this disclosure. In FIG. 8, each slot of the ODU0 includes 256 data blocks. A size of each data block is 16 bytes. For example, a 12-bit overhead region is allocated to each slot in the ODU0, for example, may be located in four high bits or four low bits of the 15th column in row 1 to row 3 of the overhead area of the ODU0; or for another example, may be located in four high bits or four low bits of the 16th column of row 1 to row 3 of the overhead area of the ODU0, as shown in FIG. 8. A TSOH represents a slot overhead region. A TSOH 1 to a TSOH 119 respectively correspond to overhead regions of the TS #1 to a TS #119.

[0126] For example, the 15th column of the overhead area of the ODU0 may further include a payload structure identifier (PSI), and the 16th column may further include a reserved (RES) bit.

[0127] In some embodiments, the overhead information may include a mapping overhead of mapping the OSU to the OTN frame, for example, increment information that is of Cm change and that is transmitted when the GMP mapping is used. When the OSU is mapped to the first service container OSTU.M, a mapping overhead of mapping the OSU to the first service container may be carried in an overhead region of a last slot included in the first service container. FIG. 9 is a schematic diagram of an overhead region of a last slot of a first service container according to an embodiment of this disclosure. CRC-4 in FIG. 9 represents a cyclic redundancy check-4. In some embodiments, the plurality of mapping overheads mapped to the first service container may also carry overhead regions of other slots included in the first service container. C1 to C6 represent quantities of OSUs in six bits. II represents an increase indication, and DI represents a decrease indication.

[0128] In some embodiments, when the bandwidth is resized, the overhead region may include interaction protocol information needed in a bandwidth resize process. When the bandwidth is increased, the interaction protocol information may be carried in an overhead region corresponding to a to-be-added slot. When the bandwidth is decreased, the interaction protocol information may be carried in an overhead region corresponding to a to-be-deleted slot. For example, the first service container includes at least M slots, and the first bandwidth corresponds to bandwidth of the at least M slots. The second service container includes M+N slots or includes M-N slots, and the second bandwidth corresponds to bandwidth of the M+N slots or bandwidth of the M-N slots. Overhead regions corresponding to N slots of the second service container in the overhead area of the OTN frame are used to carry the interaction protocol information needed for bandwidth resizing.

[0129] For example, the interaction protocol information includes at least one of the following: an RP, TSCC, a TPID, CTRL, or a TSGS.

[0130] In addition, a slot overhead region of the ODU0 shown in FIG. 8 is used as an example. 32 ODU0 frames form a multiframe. For example, five high bits of the RP, the TSCC, and the TPID may be carried in a slot overhead region of an even frame in the multiframe. Five low bits of the CTRL, the TSGS, and the TPID may be carried in a slot overhead region of an odd frame of the multiframe, as shown in FIG. 10. FIG. 10 is a schematic diagram of slot overhead regions respectively corresponding to an even multiframe and an odd multiframe according to an embodiment of this disclosure. In some scenarios, two multiframes (64 ODU0 frames) complete one transmission of the interaction protocol information.

[0131] The following describes a demapping process of each network node on a transmission path.

[0132] FIG. 11 is a schematic flowchart of a bandwidth resize method according to an embodiment of this disclosure.

The bandwidth resize method is applicable to an intermediate node or a sink node. There may be one or more intermediate nodes on a transmission path of service data. In some scenarios, the transmission path of the service data may not include the intermediate node. [0133] **1101:** Receive a higher order OTN frame, where the OTN frame is used to carry a plurality of data unit frames. [0134] **1102:** Demap a first OSU from a first service container of the OTN frame, where a payload area of the first OSU is used to carry the service data, and an overhead area of the first OSU carries a first indication that is a specified value and that indicates a bandwidth resize location of the service data in a plurality of OSUs. [0135] **1103:** Determine, based on the bandwidth resize location, a first location at which bandwidth of the plurality of OSUs changes from first bandwidth to second bandwidth.

[0136] A manner of determining the first location is described above, and details are not described herein again. [0137] **1104:** Continue to demap, from the first service container of the OTN frame, a part that is in the plurality of OSUs and that is located before the first location, and demap, from a second service container of the OTN frame, a part that is in the plurality of OSUs and that is located after the first location.

[0138] In some embodiments, determining of a start location of the second service container or a boundary point between the first service container and the second service container is described above, and is not described herein again. Based on this, step **1104** may be implemented in the following manner: the boundary point between the first service container and the second service container in the OTN frame is determined based on the first location, the first bandwidth, and the second bandwidth. Then, the part that is in the plurality of OSUs and that is located before the first location continues to be demapped from the first service container of the OTN frame based on the boundary point, and the part that is in the plurality of OSUs and that is located after the first location is demapped from the second service container of the OTN frame based on the boundary point.

[0139] In some embodiments, when a network node that performs the bandwidth resize method is a sink node, after performing step **1104**, the sink node may demap the service data from the plurality of OSUs.

[0140] In some other embodiments, when the network node that performs the bandwidth resize method is an intermediate node, after performing step **1104**, the intermediate node further continues to perform the mapping process, re-determines the first location based on the bandwidth resize location, maps the part that is in the plurality of OSUs and that is located before the first location to the first service container of the OTN frame, and maps the part that is in the plurality of OSUs and that is located after the first location to the second service container of the OTN frame. The OTN frame is sent. A process in which the intermediate node maps the OSU to the OTN is similar to a process in which the first node maps the OSU to the OTN. For example, refer to step **404** and step **405**.

[0141] The following describes in detail a procedure of a bandwidth resize method provided in embodiments of this

disclosure with reference to FIG. 12, FIG. 13A and FIG. 13B, and FIG. 14. FIG. 12 is a schematic diagram of a transmission path of service data according to an embodiment of this disclosure. A service path includes a source node NE 1, an intermediate node NE 2, and a sink node NE 3. The service path may include the intermediate node or may not include the intermediate node. In FIG. 12, that only one intermediate node is included is used as an example. FIG. 13A and FIG. 13B are schematic flowcharts of a bandwidth resize method in a scenario of increasing bandwidth according to an embodiment of this disclosure. FIG. 14 is a schematic diagram of an end-to-end bandwidth resize procedure according to an embodiment of this disclosure. Specific bandwidth resize steps may be implemented by a processor, a chip, a chip system, a module with a processing function, or the like on the network node.

[0142] Processing of the source node NE 1 includes AO processing and AI processing. In some embodiments, the AO processing may include step 1301 and step 1302. [0143] **1301:** The NE 1 maps received service data to a plurality of OSUs.

[0144] The NE 1 may receive a bandwidth resize indicate sent by a network management device or a client device, and determine that a rate of the OSU is adjusted from the first bandwidth to the second bandwidth. A rate corresponding to the first bandwidth is $R_{sub.M}$. A rate corresponding to the second bandwidth is $R_{sub.M+N}$.

[0145] Before the rate of the OSU is about to change, the NE 1 sets a first indication in the to-be-mapped OSU to a specified value. Herein, that the first indication is referred to as the BWR_IND is used as an example. A location of the BWR_IND in the OSU is x. The BWR_IND is the specified value, and indicates a bandwidth resize location y.

[0146] For ease of distinguishing, an OSU in which the first indication is set to the specified value and that is in the plurality of OSUs is referred to as a first OSU. Optionally, a second indication is further added to the first OSU. Herein, that the first indication is referred to as the HOPACC is used as an example, and $HOPACC=0$. The HOPACC records a quantity of network nodes that the OSU passes through.

[0147] Before the bandwidth resize location y in the plurality of OSUs, bandwidth is first bandwidth, and starting from the bandwidth resize location y, the bandwidth becomes second bandwidth. [0148] **1302:** The NE 1 resizes bandwidth (or adjusts a bit stream rate) of the OSU at the bandwidth resize location y. For example, after the BWR_IND, the rate of the OSU is adjusted from $R_{sub.M}$ to $R_{sub.M+N}$ at a start location of an OSU frame next to the BWR_IND. For example, the BWR_IND is carried in an OSU #i frame, and in the OSU #i frame and before the OSU #i frame, the rate of the OSU is $R_{sub.M}$, and the bandwidth of the OSU is the first bandwidth. After the OSU #i frame, the rate of the OSU is $R_{sub.M+N}$, and the bandwidth of the OSU is the second bandwidth.

[0149] The AI processing includes step 1303 and step 1304. The AI processing is used to map the OSU to an output OTN frame. For example, the OTN frame is an ODU frame herein. [0150] **1303:** Determine a first location 1, to be specific, an actual change location of the bandwidth of the bit stream of the OSU, based on the first indication. A manner of determining the first location 1 is described above, and details are not described herein again.

[0151] The NE 1 accumulates a value of the HOPACC by 1, the value of the HOPACC is 1, and the HOPACC is re-inserted into the first OSU. For example, if the value of the HOPACC is represented as HA, the actual change location of the bandwidth is $y+HA*n=y+1*n$.

[0152] In some scenarios, when the OSU does not include the second indication, in other words, the operation of accumulating the HOPACC by 1 does not need to be performed, the first location 1 is the bandwidth resize location indicated by the first indication. [0153] **1304:** Generate a mapping location Y1 corresponding to an actual change location of the bandwidth of the OSU. A manner of determining the mapping location Y1 is described above, and details are not described herein again.

[0154] **1305:** Map, based on the mapping location Y1 as the boundary point, a part that is in the plurality of OSUs and that is located before the first location 1 to a first service container (OSTU.M) of the ODU frame, and map, based on the mapping location Y1 as the boundary point, a part that is in the plurality of OSUs and that is located after the first location 1 to a second service container (OSTU.M+N) of the ODU frame.

[0155] **1306:** Send the ODU frame to the NE 2.

[0156] Processing of the intermediate node NE 2 includes B1 processing and B2 processing. The B1 processing includes demapping the OSU from the ODU frame. The B2 processing includes mapping the demapped OSU to the ODU frame.

[0157] For example, the B1 processing includes step 1307 to step 1310. [0158] **1307:** Demap the first OSU from the first service container of the received ODU frame.

[0159] The NE 2 detects the BWR_IND from the OSU carried in the received ODU frame. The BWR_IND changes from 0 to 1. Based on the BWR_IND, it is determined that the rate of the OSU is about to change at the location y. For example, the BWR_IND is detected from the first OSU.

[0160] Optionally, the value of the HOPACC is obtained from the first OSU. Herein, the value of the HOPACC is 1.

[0161] **1308:** Determine the first location 1 based on the location of the BWR_IND in the plurality of OSUs and the value of the HOPACC. The first location 1 determined by the NE 2 in the demapping process is the first location 1 used when the NE 1 performs mapping, that is, the first location is $y+HA*n=y+1*n$.

[0162] In some scenarios, when the OSU does not include the second indication, the first location 1 is the bandwidth resize location y indicated by the first indication. [0163] **1309:** Generate a demapping location Y1 corresponding to the first location 1. [0164] **1310:** Based on the demapping location Y1, continue to demap, from the first service container

(OSTU.M) of the ODU frame, the part that is in the plurality of OSUs and that is located before the first location **1**, and demap, from the second service container (OSTU.M+N) of the ODU frame, the part that is in the plurality of OSUs and that is located after the first location **1**.

[0165] For example, different demapping processes are performed based on the demapping location **Y1** as the boundary point. Before a rate change location of the OSU, an OSU before the rate change is demapped from the first service container (OSTU.M), and after the rate change location of the OSU, an OSU after the rate change is demapped from the second service container (OSTU.M+N).

[0166] For example, the **B2** processing includes step **1311** to step **1314**. The **B2** processing is used to map the OSU to the output ODU frame. [0167] **1311**: Determine a first location **2**, to be specific, the actual change location of the bandwidth of the OSU, based on the first indication. A manner of determining the first location **2** is described above, and details are not described herein again.

[0168] The **NE 1** accumulates the value of the HOPACC by **1**, the value of the HOPACC is updated to **2**, and the HOPACC is re-inserted into the first OSU. In this case, the actual change location of the bandwidth is $y+HA*n=y+2*n$. [0169] **1312**: Generate a mapping location **Y2** corresponding to the actual change location of the bandwidth of the OSU. A manner of determining the mapping location **Y2** is described above, and details are not described herein again.

[0170] In some scenarios, when the OSU does not include the second indication, in other words, the operation of accumulating the HOPACC by **1** does not need to be performed, the first location **2** is the bandwidth resize location **y** indicated by the first indication, and the mapping location **Y2** is the same as the mapping location **Y1**. [0171] **1313**: Based on the mapping location **Y2** as a boundary point, map a part that is in the plurality of OSUs and that is located before the first location **2** to the first service container (OSTU.M) of the ODU frame, and map a part that is in the plurality of OSUs and that is located after the first location **2** to the second service container (OSTU.M+N) of the ODU frame. [0172] **1314**: Send the ODU frame to the **NE 3**.

[0173] Processing of the sink node **NE 3** includes **C1** processing and **C0** processing. The **C1** processing includes demapping the OSU from the ODU frame. The **C0** processing includes demapping the service data from the OSU frame.

[0174] For example, the **C1** processing includes step **1315** to step **1318**. [0175] **1315**: Demap the first OSU from the first service container of the received ODU frame.

[0176] The **NE 3** detects the **BWR_IND** from the OSU carried in the received ODU frame. The **BWR_IND** changes from 0 to 1, and indicates that the rate of the OSU is about to change at the location **y**. For example, the **BWR_IND** is detected from the first OSU.

[0177] Optionally, the value of the HOPACC is obtained from the first OSU. Herein, the value **HA** of the HOPACC is 2. [0178] **1316**: Determine a first location **2** based on the location of the **BWR_IND** in the plurality of OSUs and the value of the HOPACC. The first location **2** determined by the **NE 2** in the demapping process is the first location **2** used when the **NE 1** performs mapping, that is, the first location **2** is $y+HA*n=y+2*n$. [0179] **1317**: Generate a demapping location **Y2** corresponding to the first location **2**.

[0180] In some scenarios, when the OSU does not include the second indication, the first location **2** is the bandwidth resize location **y** indicated by the first indication, and the mapping location **Y2** is the same as the mapping location **Y1**.

[0181] **1318**: Based on the demapping location **Y2**, continue to demap, from the first service container (OSTU.M) of the ODU frame, the part that is in the plurality of OSUs and that is located before the first location **2**, and demap, from the second service container (OSTU.M+N) of the ODU frame, the part that is in the plurality of OSUs and that is located after the first location **2**.

[0182] For example, different demapping processes are performed based on the demapping location **Y2** as the boundary point. Before the rate change location of the OSU, an OSU before the rate change is demapped from the first service container (OSTU.M), and after the rate change location of the OSU, an OSU after the rate change is demapped from the second service container (OSTU.M+N).

[0183] The **C0** processing includes step **1319**: Demap the service data from the OSU.

[0184] An embodiment of this disclosure further provides a system. The system includes a source node and a sink node, and may further include an intermediate node.

[0185] An embodiment of this disclosure further provides a bandwidth resize apparatus. The method, the apparatus, and the system are based on a same concept. The method, the apparatus, and the system have similar principles for resolving problems. Therefore, for embodiments of the apparatus and the method, refer to each other, and details are not repeated herein. The apparatus may be used in an OTN device, and the apparatus may be a processor, a chip, a chip system, a module in a processor, or the like in the OTN device. The apparatus may be implemented by a tributary board and/or a circuit board in FIG. 2. FIG. 15 is a schematic diagram of a structure of a possible bandwidth resize apparatus according to an embodiment of this disclosure. As shown in FIG. 15, the apparatus includes a processing unit **1501**, a receiving unit **1502**, and a sending unit **1503**. The processing unit **1501** is configured to: perform the bandwidth resize method according to any one of the foregoing embodiments, for example, perform a mapping operation or a demapping operation, for example, perform operations of step **402** to step **405**, and for another example,

perform operations of step **1102** to step **1104**. The receiving unit **1502** is configured to perform a receiving action of the network node in any one of the foregoing embodiments, for example, receive an OTN frame or receive service data. The sending unit **1503** is configured to perform a sending action of the network node in any one of the foregoing embodiments, for example, send the OTN frame. Optionally, the foregoing three units may further perform other related optional steps performed by the network device mentioned in any one of the foregoing embodiments, and details are not described herein again.

[0186] In this embodiment of this disclosure, unit division in this embodiment of this disclosure is an example, and is merely logical function division. In actual implementation, there may be another division manner. In addition, functional units in this embodiment of this disclosure may be integrated into one processor, or each of the units may exist alone physically, or two or more units are integrated into one unit. The integrated unit may be implemented in a form of hardware, or may be implemented in a form of a software functional unit.

[0187] FIG. **16** is a schematic diagram of a structure of another possible bandwidth resize apparatus according to an embodiment of this disclosure. As shown in FIG. **16**, an apparatus **1600** includes a communication interface **1610** and a processor **1620**. The apparatus **1600** may be used in an OTN device. The apparatus **1600** may further include a memory **1630**.

[0188] The processing unit **1501**, the receiving unit **1502**, and the sending unit **1503** shown in FIG. **15** may all be implemented by the processor **1620**. For example, the processor **1620** may be a signal processor in a line board and/or a signal processor in a tributary board shown in FIG. **2**. The processor **1620** receives and sends an OTN frame or service data by using the communication interface **1610**, and is configured to implement the method performed by the network node (the source node, the intermediate node, or the sink node) in FIG. **4**, FIG. **11**, FIG. **13A** and FIG. **13B**, or FIG. **14**. In an implementation process, each step of the processing procedure may complete the method performed by the network node in FIG. **4**, FIG. **11**, FIG. **13A** and FIG. **13B**, or FIG. **14** by using instructions in a form of software or an integrated logic circuit of hardware in the processor **1620**.

[0189] The communication interface **1610** may be a circuit, a bus, a transceiver, or any other apparatus that may be configured to exchange information. For example, the other apparatus may be a device connected to the apparatus **1600**. For example, the other apparatus may be a client device or another OTN device.

[0190] The processor **1620** may be a general-purpose processor, a digital signal processor, an application-specific integrated circuit, a field programmable gate array or another programmable logic device, a discrete gate or transistor logic device, or a discrete hardware component. The general-purpose processor may be a microprocessor, any processor, or the like. The steps of the methods disclosed with reference to embodiments of this disclosure may be directly performed by a hardware processor, or may be performed by using a combination of hardware in the processor and a software unit. Program code executed by the processor **1620** to implement the foregoing method may be stored in the memory **1630**. The memory **1630** is coupled to the processor **1620**. The coupling in this embodiment of this disclosure may be an indirect coupling or a communication connection between apparatuses, units, or modules. The coupling may be implemented in electronic, mechanical, and other forms, and is used for information exchange between the apparatuses, the units, or the modules. The processor **1620** may cooperate with the memory **1630**. The memory **1630** may be a nonvolatile memory, for example, a hard disk drive (HDD) or a solid-state drive (SSD), or may be a volatile memory, for example, a random-access memory (RAM). The memory **1630** is any other medium that can be configured to carry or store program code in a form of instructions or a data structure and that can be accessed by a computer, but is not limited thereto.

[0191] A specific connection medium between the communication interface **1610**, the processor **1620**, and the memory **1630** are not limited in this embodiment of this disclosure. In this embodiment of this disclosure, in FIG. **16**, the memory **1630**, the processor **1620**, and the communication interface **1610** are connected through a bus. The bus is represented by a thick line in FIG. **16**, and a connection manner between other components is merely described as an example, and is not limited thereto. The bus may be classified into an address bus, a data bus, a control bus, and the like. For ease of indication, the bus is indicated by using only one bold line in FIG. **16**. However, it does not indicate that there is only one bus or only one type of bus.

[0192] Based on the foregoing embodiments, an embodiment of this disclosure further provides a computer storage medium. The storage medium stores a software program, and when read and executed by one or more processors, the software program may implement the methods method provided by any one or more of the foregoing embodiments. The computer storage medium may include any medium that can store program code, for example, a Universal Serial Bus (USB) flash drive, a removable hard disk, a read-only memory (ROM), or a RAM.

[0193] Based on the foregoing embodiments, an embodiment of this disclosure further provides a chip. The chip includes a processor, configured to implement functions in any one or more of the foregoing embodiments, for example, receiving, sending, or processing the protocol frame in the foregoing method. Optionally, the chip further includes a memory, and the memory is configured to execute program instructions and data necessary for the processor. The chip may include a chip, or may include a chip and another discrete device.

[0194] An embodiment of this disclosure provides a computer-readable medium, configured to store a computer program. The computer program includes instructions used to perform the method steps in the method embodiment

corresponding to FIG. 4, FIG. 11, FIG. 13A and FIG. 13B, or FIG. 14.

[0195] Persons skilled in the art should understand that embodiments of this disclosure may be provided as a method, a system, or a computer program product. Therefore, this disclosure may use a form of a hardware-only embodiment, a software-only embodiment, or an embodiment with a combination of software and hardware. Moreover, this disclosure may use a form of a computer program product that is implemented on one or more computer-usable storage media (including but not limited to a disk memory, an optical memory, and the like) that include computer-usable program code.

[0196] This disclosure is described with reference to the flowcharts and/or block diagrams of the method, the device (system), and the computer program product according to embodiments of this disclosure. It should be understood that computer program instructions may be used to implement each process and/or each block in the flowcharts and/or the block diagrams and a combination of a process and/or a block in the flowcharts and/or the block diagrams. These computer program instructions may be provided for a general-purpose computer, a dedicated computer, an embedded processor, or a processor of another programmable data processing device to generate a machine, so that the instructions executed by a computer or the processor of another programmable data processing device generate an apparatus for implementing a specific function in one or more procedures in the flowcharts and/or in one or more blocks in the block diagrams.

[0197] It is clear that, persons skilled in the art can make various modifications and variations to this disclosure without departing from the scope of this disclosure. In this way, this disclosure is intended to cover these modifications and variations of this disclosure provided that they fall within the scope of the claims of this disclosure and their equivalent technologies.

Claims

1. A method, comprising: carrying a first indication in overheads of a first data unit frame, wherein the first indication indicates a bandwidth resize location of the first data unit frame; mapping first partial bytes that are in the first data unit frame and that are located before the bandwidth resize location to a first service container of a second data unit frame; and mapping second partial bytes that are in the first data unit frame and that are located after the bandwidth resize location to a second service container of the second data unit frame, wherein a first quantity of first slots occupied by the first service container and a second quantity of second slots occupied by the second service container are different.
2. The method of claim 1, wherein a first location to which the first indication is mapped and that is in the second data unit frame is X, wherein a second location to which a byte on the bandwidth resize location is mapped and that is in the second data unit frame is Y, and wherein Y is related to X.
3. The method of claim 1, wherein the first indication is a bandwidth resize indicate (BWR_IND).
4. The method of claim 3, wherein the BWR_IND is 1 and indicates that a rate of the first data unit frame needs to be adjusted.
5. The method of claim 1, wherein both the first data unit frame and the second data unit frame are optical transport network (OTN) frames.
6. The method of claim 1, wherein mapping the first partial bytes comprises mapping the first partial bytes to M slots of the second data unit frame, and wherein mapping the second partial bytes comprises mapping the second partial bytes to M+N slots or M-N slots of the second data unit frame.
7. The method of claim 1, wherein a first overhead area of the second data unit frame comprises a first overhead region corresponding to each slot in a payload area of the second data unit frame, and wherein a second overhead region of a last slot in the first service container comprises a mapping overhead of mapping the first data unit frame to the first service container.
8. A method, comprising: receiving a second data unit frame; demapping a first data unit frame from the second data unit frame; demapping a first indication from overheads of the first data unit frame, wherein the first indication indicates a bandwidth resize location of the first data unit frame; demapping, from a first service container of the second data unit frame, first partial bytes that are in the first data unit frame and that are located before the bandwidth resize location; and demapping, from a second service container of the second data unit frame, second partial bytes that are in the first data unit frame and that are located after the bandwidth resize location, wherein a first quantity of first slots occupied by the first service container and a second quantity of second slots occupied by the second service container in the second data unit frame are different.
9. The method of claim 8, wherein the first indication is demapped from a location X of the second data unit frame, wherein a byte on the bandwidth resize location is demapped from a location Y of the second data unit frame, and wherein Y is related to X.
10. The method of claim 8, wherein the first indication is a bandwidth resize indicate (BWR_IND).
11. The method of claim 10, wherein the BWR_IND is 1 and indicates that a rate of the first data unit frame needs to be adjusted.

- 12.** The method of claim 8, wherein both the first data unit frame and the second data unit frame are optical transport network (OTN) frames.
- 13.** The method of claim 8, wherein demapping the first partial bytes comprises demapping, from M slots of the second data unit frame, the first partial bytes, and wherein demapping the second partial bytes comprises demapping, from M+N slots or M-N slots of the second data unit frame, the second partial bytes.
- 14.** The method of claim 8, wherein a first overhead area of the second data unit frame comprises a first overhead region corresponding to each slot in a payload area of the second data unit frame, and wherein a second overhead region of a last slot in the first service container comprises a mapping overhead of mapping the first data unit frame to the first service container.
- 15.** An apparatus, comprising: a memory configured to store instructions; and one or more processors coupled to the memory and configured to execute the instructions to cause the apparatus to: carry a first indication in overheads of a first data unit frame, wherein the first indication indicates a bandwidth resize location of the first data unit frame, and wherein the first indication indicates a bandwidth resize location of the first data unit frame; map first partial bytes that are in the first data unit frame and that are located before the bandwidth resize location to a first service container of a second data unit frame; and map second partial bytes that are in the first data unit frame and that are located after the bandwidth resize location to a second service container of the second data unit frame, wherein a first quantity of first slots occupied by the first service container and a second quantity of second slots occupied by the second service container are different.
- 16.** The apparatus of claim 15, wherein a first location to which the first indication is mapped and that is on the second data unit frame is X, wherein a second location to which a byte on the bandwidth resize location is mapped and that is on the second data unit frame is Y, and wherein Y is related to X.
- 17.** The apparatus of claim 15, wherein the first indication is a bandwidth resize indicate (BWR_IND).
- 18.** The apparatus of claim 17, wherein the BWR_IND is 1 and indicates that a rate of the first data unit frame needs to be adjusted.
- 19.** The apparatus of claim 15, wherein both the first data unit frame and the second data unit frame are optical transport network (OTN) frames.
- 20.** The apparatus of claim 15, wherein the one or more processors are further configured to execute the instructions to cause the apparatus to: map the first partial bytes by mapping the first partial bytes to M slots of the second data unit frame; and map the second partial bytes by mapping the second partial bytes to M+N slots or M-N slots of the second data unit frame.
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